

A new *Laccaria* species from cloud forest of Fortuna, Panama

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Abstract A new species of Agaricales, called *Laccaria stellata*, was collected in a premontane cloud forest in the Fortuna Forest Reserve, Panama, and it is described based on morphological and molecular characteristics. It differs morphologically from all known species of *Laccaria* by minute, pinkish-orange colored basidiomata, a very thin and translucent pileus, very distant lamellae, 4-spored basidia, and globose basidiospores covered by relatively large echinulae. Molecular rDNA sequence data confirm the separation of this new species from other *Laccaria* species for which rDNA sequence data are available.

Keywords Basidiomycota · Taxonomy · Phylogeny · *Laccaria* · Hydnangiaceae · Panama

Introduction

The species of *Laccaria* Berk. & Broome (Hydnangiaceae, Agaricales, Basidiomycota) play important roles as ectomycorrhizal partners of trees throughout forest ecosystems worldwide (McNabb 1972; Mueller 1992; Nara et al. 2003; Roy et al. 2008). The species within the genus show

different host ranges, e.g., *Laccaria amethystina* Cooke has a wide host range, whereas *Laccaria maritima* (Theodor.) Singer ex Huhtinen is specifically associated with species of the genus *Pinus* (Huhtinen 1987; Roy et al. 2008).

Laccaria spp. morphologically differ from species of other genera of Agaricales by a combination of globose, ellipsoid to oblong echinulate basidiospores, collybioid to omphaloid habit and orange to brownish or violet colored carpophores, a few species showing grey to black or whitish coloration (Berkeley and Broome 1883; Imai 1938; Mueller 1992; Vellinga 1986; Wang et al. 2004). The morphological discrimination of the species within the genus, however, can be very difficult, and cryptic speciation has occurred, as shown by Sheedy et al. (2013).

The new species *Laccaria stellata* was repeatedly sampled in premontane cloud forest of the Fortuna Forest Reserve in Western Panama, Mesoamerica, which belongs to an area recognized and characterized by Myers et al. (2000) as one of 25 biodiversity hotspots worldwide. The Fortuna Forest Reserve is located in the Cordillera de Talamanca Gualaca district, province of Chiriquí in Panama and forms the southeastern border of the protected areas of Palo Seco Forest, La Amistad and the Volcán Barú. The National Park La Amistad extends into Costa Rica and includes the largest continuous forest area in Central America. The Fortuna Forest Reserve comprises 19,500 ha with altitudes ranging from 1,000 m to 2,238 m a.s.l., with an annual precipitation of 4,000 mm and an average annual temperature between 16 and 20 °C (Cavelier et al. 1997; Araúz 2007).

The forest is dominated by trees of Fagaceae and Juglandaceae and a high number of fruiting bodies of ectomycorrhizal fungi (e.g., species of *Amanita*, *Cantharellus*, *Cortinarius*, *Laccaria*, *Lactarius*, *Russula*) were observed during fieldwork (Castillo Jiménez 2013). In the present study, we describe the new species *Laccaria stellata* based on

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Table 1 Species names, location of the examined specimen, voucher data, and accession numbers of ITS sequences included in this study. They belong to *Laccaria* spp. (Hydnangiaceae), except for *Mythicomyces corneipes*, which is used as outgroup (Psathyrellaceae)

Species	Location	Voucher specimen	Genbank accession no.	Mycobank accession no.	Reference
<i>Laccaria acanthospora</i>	China	—	JX504161.1	—	Wilson et al. 2013
<i>Laccaria acanthospora</i>	China	—	JX504119.1	—	Wilson et al. 2013
<i>Laccaria acanthospora</i> TYPE	China	—	JX504102.1	—	Wilson et al. 2013
<i>Laccaria alba</i>	China	—	JX504126.1	MB371126	Wilson et al. 2013
<i>Laccaria alba</i>	China	—	JX504131.1	MB371126	Wilson et al. 2013
<i>Laccaria alba</i>	China	—	JX504129.1	MB371126	Wilson et al. 2013
<i>Laccaria amethysteo-occidentalis</i>	North America	—	HQ650762.1	MB114139	Kranabetter et al. 2012
<i>Laccaria amethysteo-occidentalis</i>	North America	—	JX504112.1	MB114139	Wilson et al. 2013
<i>Laccaria angustilamella</i>	China	—	JX504132.1	MB371125	Wilson et al. 2013
<i>Laccaria angustilamella</i>	China	—	JX504118.1	MB371125	Wilson et al. 2013
<i>Laccaria aurantia</i>	China, Yunnan	KUN-F 78557	JQ670895.1	MB564443	Popa et al. 2014
<i>Laccaria aurantia</i>	China, Yunnan	MB-FB-101109	JQ681209.1	MB564443	Popa et al. 2014
<i>Laccaria bicolor</i>	North America	—	JX504116.1	MB332790	Wilson et al. 2013
<i>Laccaria bicolor</i>	North America	—	JX504115.1	MB332790	Wilson et al. 2013
<i>Laccaria bullipellis</i> TYPE	China	—	JX504100.1	—	Wilson et al. 2013
<i>Laccaria canaliculata</i>	Australia	—	JX504136.1	MB100701	Wilson et al. 2013
<i>Laccaria</i> cf. <i>canaliculata</i>	Australia	—	JX270720.1	MB100701	Sheedy et al. 2013
<i>Laccaria</i> cf. <i>lateritia</i>	Australia	—	JX270736.1	MB530053	Sheedy et al. 2013
<i>Laccaria</i> cf. <i>lateritia</i>	Australia	—	JX270695.1	MB530053	Sheedy et al. 2013
<i>Laccaria</i> cf. <i>masoniae</i>	Australia	—	JX270741.1	MB530053	Sheedy et al. 2013
<i>Laccaria fulvogrisea</i>	China, Yunnan	KUN-F 78556	JQ670896.1	MB564442	Popa et al. 2014
<i>Laccaria fulvogrisea</i>	China, Yunnan	MB-FB-101105	JQ681210.1	MB564442	Popa et al. 2014
<i>Laccaria himalayensis</i>	China	—	JX504104.1	—	Wilson et al. 2013
<i>Laccaria himalayensis</i> TYPE	China	—	JX504101.1	—	Wilson et al. 2013
<i>Laccaria laccata</i>	Europe	—	UDB000106	MB119173	UNITE
<i>Laccaria laccata</i>	Europe	—	UDB000105	MB119173	UNITE
<i>Laccaria masoniae</i>	Australia	—	JX504139.1	MB530053	Wilson et al. 2013
<i>Laccaria montana</i>	North America	—	DQ149865.1	MB316189	Osmundson et al. 2005
<i>Laccaria montana</i>	North America	—	DQ149862.1	MB316189	Osmundson et al. 2005
<i>Laccaria murina</i>	Japan	—	AB211271.1	MB271842	Nara 2006
<i>Laccaria nobilis</i>	North America	—	DQ149863.1	MB106943	Osmundson et al. 2005
<i>Laccaria nobilis</i>	North America	—	DQ149867.1	MB106943	Osmundson et al. 2005
<i>Laccaria ochropurpurea</i>	North America	—	JX504169.1	MB295607	Wilson et al. 2013
<i>Laccaria ohiensis</i>	North America	—	JX504106.1	MB287397	Wilson et al. 2013
<i>Laccaria proxima</i>	North America	—	DQ149852.1	MB438537	Osmundson et al. 2005
<i>Laccaria proxima</i>	North America	—	JX504142.1	MB438537	Wilson et al. 2013
<i>Laccaria proxima</i>	Europe	—	AY750156.1	MB438537	Cline et al. 2005
<i>Laccaria proxima</i>	Europe	—	DQ499641.1	MB438537	Krueger & Gargas 2006, unpublished
<i>Laccaria proxima</i>	North America	—	DQ149852.1	MB438537	Osmundson et al. 2005
<i>Laccaria pseudomontana</i>	North America	—	DQ149871.1	MB500895	Osmundson et al. 2005
<i>Laccaria pseudomontana</i>	North America	—	DQ149870.1	MB500895	Osmundson et al. 2005
<i>Laccaria pumila</i>	North America	—	DQ149873.1	MB248633	Osmundson et al. 2005
<i>Laccaria pumila</i>	North America	—	DQ149864.1	MB248633	Osmundson et al. 2005
<i>Laccaria salmonicolor</i> TYPE	China	—	JX504143.1	—	Wilson et al. 2013

Table 1 (continued)

Species	Location	Voucher specimen	Genbank accession no.	Mycobank accession no.	Reference
<i>Laccaria stellata</i>	Panama	MB-002396	KP877340	MB811780	present publication
<i>Laccaria stellata</i>	Panama	MB-002397	KP877339	MB811780	present publication
<i>Laccaria tortilis</i>	North America	–	DQ149872.1	MB438538	Osmundson et al. 2005
<i>Laccaria tortilis</i>	Europe	–	UDB001589	MB438538	UNITE
<i>Laccaria vinaceoavellanea</i>	Japan	–	JN942785.1	MB316190	Schoch et al. 2012
<i>Laccaria vinaceoavellanea</i>	Japan	–	JN942803.1	MB316190	Hosaka 2011, unpublished
<i>Laccaria yunnanensis</i>	China, Yunnan	KUN-F 78558	JQ670897.1	MB564444	Popa et al. 2014
<i>Laccaria yunnanensis</i>	China, Yunnan	MB-FB-101108	JQ681208.1	MB564444	Popa et al. 2014
<i>Mythicomycetes corneipes</i>	Germany	–	KC964108.1	MB563788	Strittmatter & Oberbauer 2013, unpublished
<i>Mythicomycetes corneipes</i>	Germany	–	DQ404393.1	MB563788	Matheny & Hibbett 2006, unpublished

macromorphological and micromorphological characteristics and illustrations, as well as molecular sequence data.

Material and methods

Collection and field site

The fruiting bodies were collected in the period of November 2011 to August 2012 by SY. Castillo Jiménez (2013). All collections were sampled in close vicinity of several trees of *Quercus insignis* M. Martens & Galeotti, coded as Q1, Q2, Q3, Q4, and Q5. The locations of the collections were recorded using a GPSMAP 62s (Garmin Deutschland GmbH, Garching, Germany).

The collections were documented and dried on silica gel. The holotype collection is deposited in the National Herbarium of the University of Panama (PMA) of the Universidad Autónoma de Chiriquí (UCH) at David, Panamá. The Isotype is deposited in the Herbarium Marburgense (MB). Paratypes are deposited in the Herbarium of the Universidad Autónoma de Chiriquí (UCH) at David, Panamá, Herbarium Marburgense (MB), University of Marburg, Germany and Herbarium Senckenbergianum (FR), Frankfurt, Germany.

Light microscopy

The material was analyzed by light microscopy and structures were measured in 5 % KOH, water, or in a solution of lactic acid (90 %) / glycerol (10 %) using a Laborlux D microscope (Leica, Wetzlar, Germany). Up to 20 basidiospores, basidia, and cystidia were observed for each sample. The basidiospore dimensions were measured excluding ornamentation and the apiculus. In addition, the length / width ratio (= Q) was calculated.

Scanning electron microscopy

For scanning electron microscopy (SEM), fragments of the specimen SYC-122 (dried on silica gel) were fixed onto sample plates using carbon glue. The samples were coated twice for 60 sec with gold with an agar sputter coater. The samples were observed and documented by digital photos with a Hitachi S 4500 and a Hitachi S530 (Hitachi, Japan) scanning electron microscope.

DNA extraction, PCR amplification, DNA sequencing

Ribosomal nuclear DNA was isolated from fragments of a fruiting body of the specimens SCY 109 (Q3, 24.1.2014) and SCY 207 (Q2, 26.1.2014). The fungal tissue SCY 109 was crushed in a mixer mill (Retsch, model MM 301) together with tungsten carbide (2.5 mm diam.) for 30 sec min at 30 Hz. The genomic DNA was isolated and cleaned following the CTAB method described by Talbot (2001) and modified by J. Weisenborn. The ITS region including ITS1, the 5.8S rRNA gene, and ITS2 was amplified with the primers ITS1-F and ITS4 (Gardes and Bruns 1993; White et al. 1990), peqGOLD dNTPs, and peqGOLD Taq using a thermocycler Primus 25. PCR products were purified using the PEQLAB E.Z.N.A.® peg Gold Cycle Pure kit following the protocol of the manufacturer. Sequences were obtained at the BiK-F sequencing facility at Frankfurt am Main, Germany. The treatment of the SCY 207 specimen differed from the treatment of the SCY 109 specimen by a nested polymerase chain reaction (PCR) with the primer combination NSI1 and NLB4 (Kendall and Rygiewicz 2005) for the first reaction, and the primers ITS1F (Gardes and Bruns 1993) and ITS4 (White et al. 1990) for the second reaction. The amplification with the first primer combination was carried out in a peqSTAR 96 Universal (PEQLAB Biotechnologie GmbH, Erlangen,

Fig. 1 Basidiomata of *Laccaria stellata* [MB-002395] (scale bar = 2 cm)



Germany). Purification of the samples was performed using a PCR purification kit (Invitex, Berlin, Germany). The purified samples were sent to LGC, Berlin, Germany for sequencing.

Phylogenetic analyses

Consensus sequences of the new species and sequences retrieved from GenBank (Benson et al. 2005) and UNITE (Abarenkov et al. 2010) (Table 1) were aligned and edited

with CodonCode Aligner (CodonCode Aligner version 5.0.1). First the sequence alignments were computed using Mafft (Kato and Standley 2013); later, we performed sequence alignments using the ClustalW algorithm (FFT-NS-1 algorithm). The sequence alignment was corrected using the Gblocks tool version 0.91b (Castresana 2000), allowing smaller final blocks, gap positions within the final blocks as well as less strict flanking positions. The final data set contained a total of 54 sequences with 604 positions.

Fig. 2 *Laccaria stellata* [MB-002395] **a.** Basidioma. **b.** Pileipellis. **c.** Basidiospores **d.** Hymenium with basidia and pleurocystidia

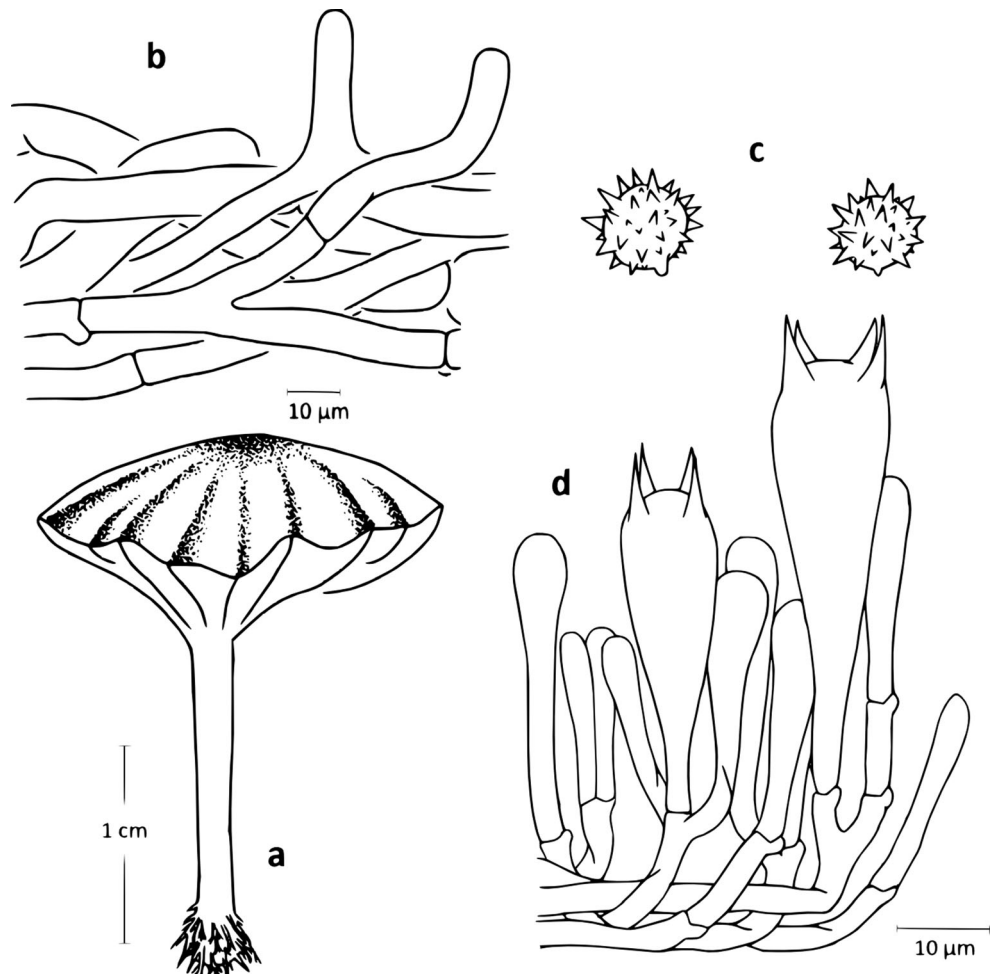
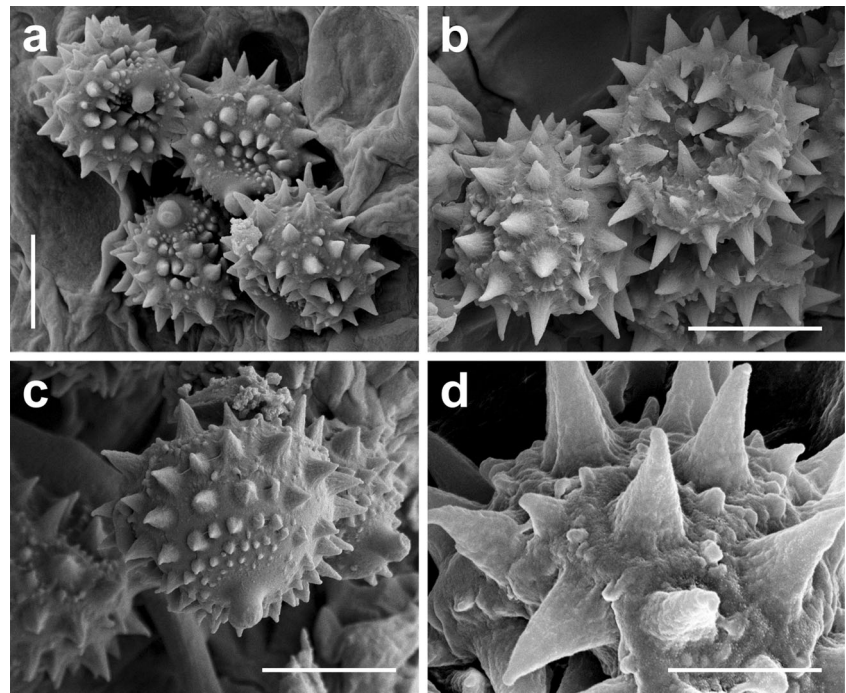


Fig. 3 a–c. SEM's of *Laccaria stellata* basidiospores Scale bars = 5 µm d. Detail of the ornamentation of a basidiospore, Scale bar = 2 µm [FR-0219864]



The Phylogenetic trees were calculated using Maximum likelihood and Bayesian analyses. Maximum likelihood analyses were performed by RAxML-HPG BlackBox (8.0.0) using a GTRCAT model through the CIPRES Science Gateway (Stamatakis et al. 2005, 2008; Stamatakis 2006; Miller

et al. 2010). Bootstrap values and Bayesian probability values below 50 are not shown. Maximum Likelihood branch support was measured using nonparametric bootstrapping (Felsenstein 1985) with 1000 replicates. Bayesian analysis was performed using MrBayes 3.2.2 (Huelsenbeck and

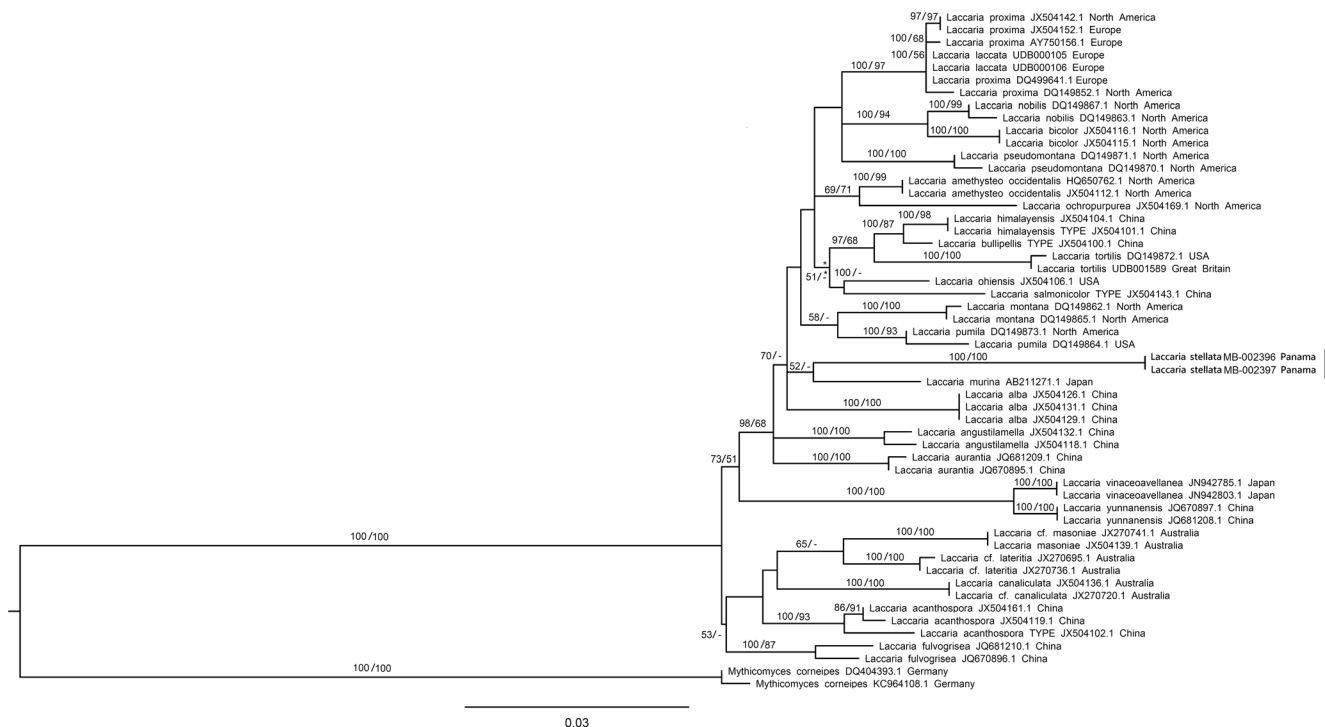


Fig. 4 Maximum Likelihood tree based on ITS 1, 5.8S rDNA and ITS 2 sequences of the new *Laccaria* species from Panama and selected *Laccaria* sequences from GenBank and UNITE. The number on the left

indicates the Bayesian analysis posterior probability values in percent. The number on the right denotes the bootstrap value (1000 replicates) of the Maximum Likelihood tree

Ronquist 2001; Ronquist and Huelsenbeck 2003; Ronquist et al. 2012) using the Cipres Science Gateway (Miller et al. 2010). The Bayesian analysis included two independent runs of 10 million generations with four chains per run, a sampling frequency of 1000, a print frequency of 1000 generations and a burn-in of 25 % with $nst = 1$, $rates = +G$, and default settings for the other parameters were used. The final Maximum Likelihood consensus tree is the result of the 50 % majority rule plus all compatible groups from both analyses.

Results

Taxonomy

Laccaria stellata F. Popa, S.Y. Castillo, sp. nov.

Diagnosis: The new species differs from known species of *Laccaria* by the following combination of characteristics: Basidiomata minute to small; Pileus 1.4–2.2 cm in diameter, pinkish-orange to flesh-colored, translucently striate almost to the center; Lamellae distant, adnate to decurrent, Basidia 4-spored; Basidiospores (7) 7.5–8.5 (9) $\times$ 7.5–9 (9.5) μm , $Q = 0.95\text{--}1.0\text{--}1.1$, globose, echinulate, echinulae (1.5) 2.5–3 (3.3) μm high.

Holotypus: Panama, Chiriquí province, Gualaca, Reserva Forestal Fortuna, Quebrada Honda A, 08°45'12.1" N, 082°14'21.9" W, at approx. 1199 m a.s.l., close to *Quercus insignis* (Q3), 26.12.2011, Castillo Jimenéz, SY., SCY-106 (PMA). *Isotype* at MB [MB-002395], *Paratypes* are in MB, FR and UCH. Herbarium Marburgense number: MB-002395, Mycobank accession number: MB 811780; GenBank accession number: KP877340

Basidiomata minute to small. Pileus 1.4–2.2 cm in diameter, pinkish-orange to flesh-colored [7.5R 7/10–7.5R 7/12, 10R 6/12–10R 6/14] (Long and Luke 2001), first broadly conical then plano-convex, with age center can be depressed, translucently striate almost to the center, surface sometimes with radial ridges, smooth, shiny, fibrillose, slightly hygrophanous, margin very thin, wavy. Taste and smell not observed. Lamellae 0.1 cm thick, very distant, adnate to slightly decurrent, ventricose, first whitish then pinkish to orange. Stipe 1.5–2.5 $\times$ 0.2 cm, cylindrical, fistulose, longitudinally fibrillose, slightly bulbous towards the base, concolorous with the pileus, with age more whitish to the base. Basal mycelium can be strongly developed, whitish.

Basidia (35) 37–43 (45) $\times$ (9) 10 (12) μm , clavate, hyaline, 4-spored, sterigmata up to 9 μm long, straight to slightly bent. Basidiospores (7) 7.5–8.5 (9) $\times$ 7.5–9 (9.5) μm , $Q = 0.95\text{--}1.0\text{--}1.1$, globose, hyaline, echinulate, echinulae (1.5) 2.5–3 (3.3) μm high, each surrounded by a ring of small warts evident by SEM, hilar appendix 1.5–2 μm long, coloration of the spore print was not observed. Pleurocystidia and cheilocystidia present, numerous, similar, clavate, cylindrical to capitate, (20) 25–40 (42) $\times$ (3) 5–7 (7.5) μm , sometimes with a short basal

cell. Cheilocystidia intermixed with basidia. Stipe cortex and trama composed of longitudinally arranged elongated hyphae, which increase up to 20 μm in diameter towards the center of the stipe. Caulocystidia absent. Pileotrama less than 1 mm in thickness, formed by interwoven hyphae without fasciculate bundles arising. Pileipellis a cutis consisting of simple hyphae intermixed with radiating, interwoven, thin-walled hyphae with intracellular yellowish to bright brownish coloration, inflated up to 20 μm diam. Clamp connections present at most septa.

Etymology: Because of the thin pileotrama, the distant lamellae appear star-shaped through the pileus.

Geographic distribution / host range: The new species has been observed and repeatedly collected close to roots of *Quercus insignis* trees in premontane rain forest in the area called Quebrada Honda A in the Reserva Forestal Fortuna, Gualaca district, Chiriquí province, Panama. This forest is about 300 years old and with approx. 25 % basal area dominated by old ectomycorrhizal trees of *Quercus insignis* (Fagaceae) and *Oreomunnea mexicana* (Standl.) J.-F. Leroy (Juglandaceae).

Material examined: PANAMA: Chiriquí province, Gualaca district, Reserva Forestal Fortuna, Quebrada Honda A. *Quercus insignis* **Q1** (8°45'09.9" N; 82°14'21.3" W), at approximately 1189 m, **SYC-108**, *Paratype* [MB-002446] (26.1.2012); *Q. insignis* **Q2** (8°45'10.0" N, 82°14'21.8" W) at approximately 1181 m, **SYC-207**, *Paratype* [MB-002397] (26.01.2012), **SYC-122**, *Paratype* [FR-0219864] (26.04.2012), **SYC-137**, *Paratype* [FR-0219865] (26.05.2012), **SYC-196**, *Paratype* [UCH] (26.08.2012); *Q. insignis* **Q3** (8°45'12.1" N, 82°14'21.9" W) at approximately 1199 m, **SYC-106**, *Holotype* [PMA], *Isotype* [MB-002395] (26.12.2011), **SYC-109** [MB-002396] (26.01.2012), **SYC-120**, *Paratype* [UCH] (26.3.2012), **SYC-172**, *Paratype* [UCH] (26.06.2012); *Q. insignis* **Q4** (8°45'09.7" N, 82°14'23.4" W) at approximately 1191 m, **SYC-205**, *Paratype* [UCH] (26.08.2012), *Q. insignis* **Q5** (8°45'12.6" N, 82°14'24.4" W) at approximately 1205 m, **SYC-130**, *Paratype* [UCH] (26.04.2012), **SYC-156**, *Paratype* [UCH] (26.05.2012), *Q. insignis* **Q3 SYC 209**, *Paratype* [MB-002491] (24.01.2014).

Discussion: *Laccaria stellata* is characterized by orange to flesh-colored, minute basidiomata with broadly convex to plane, very thin and translucent pilei, so the distant lamellae are visible from above the thin pileotrama. Basidia are 4-spored and develop globose basidiospores ornamented by echinulae that are rather large in relation to the size of the basidiospores.

Based on characteristics like small, orange to flesh-colored fruiting bodies and globose basidiospores, this species can be placed in the section *Laccata* stirps *Ohiensis* (Bon 1983), but analyses by Mueller (1992) as well as molecular sequence data show that the stirps is polyphyletic. *Laccaria stellata*

differs from other species within this stirps, like *L. lateritia* Malençon, *L. ohiensis* (Mont.) Singer, *L. pumila* Fayod, *L. striatula* (Peck) Peck and *L. tortilis* (Bolton) Cooke by the morphological combination of small fruiting body size, the very thin pileotrama without fasciculate bundle of hyphae, very distant lamellae, as well as the combination of 4-spored basidia, small basidiospores with prominent echinulation. *Laccaria pumila*, *L. lateritia* and *L. tortilis* can be very well distinguished by having 2-spored basidia compared to *L. stellata*. *L. striatula* differ by larger fruiting bodies, without cystidia and a different ecology. *L. ohiensis* is morphologically very similar to *L. stellata*, but the new described species differs by smaller delicate fruiting bodies, slightly decurrent and very distant lamellae, missing fasciculate bundles of hyphae, capitate cheilocystidia and pleurocystidia and globose spores, as well as its unique ITS sequence.

The phylogeny presented in Figs. 1, 2, 3 and 4 confirms that the stirps *Ohiensis* is polyphyletic, and it shows that the molecular sequence of the rDNA region of *L. stellata* differs from the corresponding sequences known from other *Laccaria* species. According to sequence data, *L. stellata* is most closely related to *L. murina* S. Imai from Japan, but the low bootstrap values and the long branch indicate only weak support for this relationship. Additionally, *L. murina* differs morphologically from *L. stellata* by greyish colors and less prominently echinulate spores.

All available collections of the new species *L. stellata* were sampled close to *Quercus insignis* trees, in most cases close to the base of the stem, within the Fortuna Forest Reserve. Apparently, this species has a small ecological amplitude and its occurrence might be restricted to *Quercus insignis* that is distributed in Mesoamerica and Mexico (cited as *Q. oocarpa*, Muller 1960). The cloud forest of the Fortuna Forest Reserve has a high ecological and socio-economic importance due to a high number of endemic species of plants and ecosystem services for the country of Panama (Mendieta 2005). However, almost nothing is known about the ectomycorrhizal fungi and their symbiosis with the tree species in this area. The Fortuna Forest Reserve managed by the indigenous Ngöbe-Buglé communities is threatened by environmental degradation, mining, and construction of dams. By the destruction of habitats, many species of fungi will disappear without ever being known to science, including information concerning their functions in the ecosystem, or possible benefits for humans.

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